OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 37 OF 55

AI-BASED TRAFFIC VIOLATION DETECTION AND AUTOMATED LICENSE PLATE RECOGNITION SYSTEM

Domain: Intelligent Transportation & ANPR Surveillance | **Module:** project_37_traffic_violation

OFFICIAL ABSTRACT

As urban vehicle density continues to grow, enforcing traffic rules and maintaining road order have become increasingly challenging for law enforcement agencies. Traditional manual monitoring is insufficient for continuous surveillance and prone to human error. To address this, this study proposes an AI-based traffic violation detection and automated license plate recognition (ALPR) system. The system integrates advanced computer vision and deep learning algorithms to automatically monitor traffic cameras and detect common traffic infractions, such as red-light violations, speeding, illegal lane changes, and wrong-way driving. Object detection and tracking models, such as YOLO and DeepSORT, are utilized to identify vehicles, monitor their trajectories, and detect anomalous driving behaviors in real-time. Upon detecting a violation, the system automatically isolates the vehicle's region of interest and applies a dedicated deep learning-based ALPR model to extract the license plate characters under diverse environmental conditions, including poor lighting and motion blur. The system is validated on traffic video datasets, demonstrating high accuracy in both violation detection and license plate transcription. The recognized license plates are then matched against a database to generate automated citations or alert authorities. The deployment of this end-to-end automated solution improves enforcement efficiency, reduces the burden on manual surveillance, and acts as a strong deterrent against traffic violations, ultimately promoting safer driving behavior.

📌 Project Summary & System Specifications

Project Title	AI-BASED TRAFFIC VIOLATION DETECTION AND AUTOMATED LICENSE PLATE RECOGNITION SYSTEM
Specialization / Category	Intelligent Transportation & ANPR Surveillance

Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5037`
Local Implementation	`project_37_traffic_violation`